

ULIASTAI, Mongolia

WMO: 442720

Lat: 47.7272N

Lon: 96.8475E

Elev: 1759

StdP: 81.90

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.7	-32.7	-38.0	0.1	-34.3	-36.1	0.2	-32.2	6.5	-14.7	5.2	-19.0	1.0	180	0.660

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	28.0	14.6	26.1	13.9	24.4	13.0	16.1	24.9	15.0	23.6	14.0	22.1	3.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.2	11.7	17.8	11.8	10.7	17.8	10.6	9.8	17.5	52.2	24.6	48.5	23.5	45.3	22.2	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.4	6.3	5.3	-37.3	31.0	2.7	1.9	-39.3	32.4	-40.9	33.5	-42.4	34.6	-44.4	36.0		
(5)			-37.3	17.6	2.7	1.8	-39.3	18.9	-40.9	19.9	-42.4	20.9	-44.4	22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-22.8	-18.5	-8.9	2.6	8.7	14.9	17.3	14.9	8.3	-0.2	-11.6	-19.8	(6)
(7)		DBStd	14.82	5.32	5.91	6.82	4.98	4.42	3.55	3.15	3.35	3.97	5.01	6.46	4.98	(7)
(8)		HDD10.0	4674	1017	798	587	223	77	6	0	3	76	315	649	923	(8)
(9)		HDD18.3	7161	1275	1032	846	471	299	112	58	113	301	573	899	1181	(9)
(10)		CDD10.0	597	0	0	0	2	36	152	226	156	26	0	0	0	(10)
(11)		CDD18.3	41	0	0	0	0	0	8	25	8	0	0	0	0	(11)
(12)		CDH23.3	599	0	0	0	0	12	137	322	126	3	0	0	0	(12)
(13)		CDH26.7	117	0	0	0	0	0	19	78	19	0	0	0	0	(13)
(14)	Wind	WSAvg	1.6	0.9	1.0	1.5	2.1	2.4	2.1	1.9	1.8	1.8	1.5	1.1	1.0	(14)
(15)	Precipitation	PrecAvg	199	3	2	5	9	15	32	59	43	20	9	4	3	(15)
(16)		PrecMax	327	11	11	21	32	47	90	178	93	76	30	15	9	(16)
(17)		PrecMin	90	0	0	0	0	0	4	9	0	1	0	0	0	(17)
(18)		PrecStd	54	2	2	4	7	12	19	35	24	15	8	3	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.1	-1.1	10.3	19.0	24.9	29.1	31.5	28.8	23.3	15.5	6.7	-3.5	(19)
(20)			MCWB	-8.1	-4.0	2.8	8.5	11.7	14.6	16.8	14.8	11.1	6.5	1.0	-5.9	(20)
(21)		2%	DB	-10.0	-4.7	6.7	16.2	22.5	26.6	28.8	26.5	20.7	12.7	3.5	-8.0	(21)
(22)			MCWB	-11.2	-7.0	0.6	6.7	10.8	13.8	15.4	13.8	10.4	4.9	-1.1	-9.4	(22)
(23)		5%	DB	-12.6	-7.2	4.1	13.7	20.2	24.9	26.8	24.5	18.6	10.7	0.6	-10.2	(23)
(24)			MCWB	-13.7	-9.0	-0.8	5.4	9.5	13.1	14.2	13.3	9.1	3.6	-3.0	-11.4	(24)
(25)		10%	DB	-14.9	-9.4	1.5	11.5	17.9	23.2	24.9	22.6	16.5	8.3	-2.2	-12.3	(25)
(26)			MCWB	-15.9	-10.9	-2.3	3.9	7.9	12.2	13.5	12.5	8.3	2.3	-4.9	-13.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.9	-3.8	3.1	8.9	13.4	16.7	18.5	16.3	12.6	7.2	1.6	-6.0	(27)
(28)			MCDWB	-6.3	-1.2	9.8	16.9	23.9	26.2	26.4	25.1	20.6	14.4	6.2	-3.8	(28)
(29)		2%	WB	-11.1	-6.9	0.9	7.2	11.2	14.9	16.9	15.1	11.0	5.4	-0.9	-9.3	(29)
(30)			MCDWB	-10.0	-4.8	6.0	15.2	20.7	24.6	26.3	23.1	18.6	11.8	3.2	-8.1	(30)
(31)		5%	WB	-13.6	-9.0	-0.6	5.7	9.8	13.8	15.7	14.1	9.9	4.1	-2.8	-11.4	(31)
(32)			MCDWB	-12.5	-7.2	3.9	13.0	19.0	22.8	24.1	21.8	17.1	10.2	0.6	-10.3	(32)
(33)		10%	WB	-15.8	-10.9	-2.2	4.2	8.5	12.8	14.6	13.2	8.7	2.7	-4.7	-13.4	(33)
(34)			MCDWB	-14.9	-9.4	1.6	10.8	17.1	21.0	22.7	20.9	15.8	8.3	-2.1	-12.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.2	12.7	14.1	11.9	12.9	12.6	11.7	12.2	13.5	13.1	11.1	9.4	(35)
(36)			MCDBR	11.5	13.7	14.0	15.1	16.5	15.4	15.2	15.4	16.3	16.2	14.1	10.7	(36)
(37)			MCWBR	10.4	11.9	9.6	8.6	9.1	7.4	6.6	6.9	8.7	10.0	10.1	9.6	(37)
(38)		5% WB	MCDWB	11.4	13.7	13.6	14.2	15.4	13.8	12.8	13.1	14.4	15.5	13.6	10.6	(38)
(39)			MCWBR	10.3	11.9	9.5	8.5	9.2	7.2	6.6	6.4	8.1	9.8	9.9	9.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.225	0.253	0.296	0.339	0.338	0.349	0.348	0.326	0.292	0.274	0.248	0.229	(40)	
(41)		taud	2.336	2.268	2.209	2.232	2.294	2.353	2.390	2.424	2.473	2.438	2.380	2.325	(41)	
(42)		Ebn at Noon	886	922	923	911	923	912	907	916	924	888	842	829	(42)	
(43)		Edh at Noon	68	93	118	128	126	120	114	105	90	78	66	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.77	2.85	4.41	5.74	6.74	6.77	6.42	5.69	4.78	3.11	1.79	1.39	(44)	
(45)		RadStd	0.14	0.18	0.16	0.24	0.41	0.33	0.39	0.25	0.21	0.20	0.14	0.10	(45)	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(47) Regional (5 neighbors)		N/A	N/A	-1.95	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page